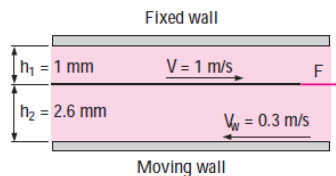


Fluid properties: Practice Problems

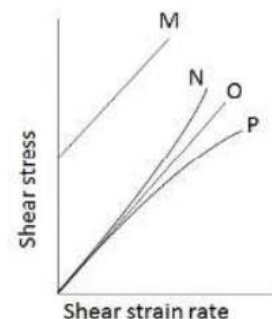
Viscosity

1. What is viscosity? What is the cause of it in liquids and in gases? Do liquids or gases have higher viscosities?
2. What is a Newtonian fluid? Is water a Newtonian fluid?
3. Consider two identical small glass balls dropped into two identical containers, one filled with water and the other with oil. Which ball will reach the bottom of the container first? Why?
4. How does the viscosity of (a) liquids and (b) gases vary with temperature?
5. A thin 20-cm x 20-cm flat plate is pulled at 1 m/s horizontally through a 3.6-mm-thick oil layer sandwiched between two plates, one stationary and the other moving at a constant velocity of 0.3 m/s, as shown in Fig. The dynamic viscosity of oil is 0.027 Pas. Assuming the velocity in each oil layer to vary linearly, (a) plot the velocity profile and find the location where the oil velocity is zero and (b) determine the force that needs to be applied on the plates to maintain this motion.

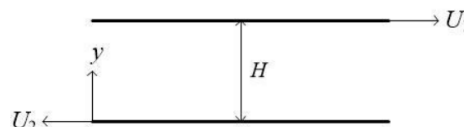


6. The viscosity of a fluid is to be measured by a viscometer constructed of two 3-ft-long concentric cylinders. The inner diameter of the outer cylinder is 6 in, and the gap between the two cylinders is 0.05 in. The outer cylinder is rotated at 250 rpm, and the torque is measured to be 1.2 lbf. ft. Determine the viscosity of the fluid. Answer: 0.000648 lb.s/ft²
7. Rheological diagram of different types of fluids is shown in figure. Column I represents the nature of the fluid and column II represents the curve showing the variation of shear stress against shear strain rate.

Column I	Column II
Newtonian	M
Shear thinning	N
Shear thickening	O
Bingham plastic	P



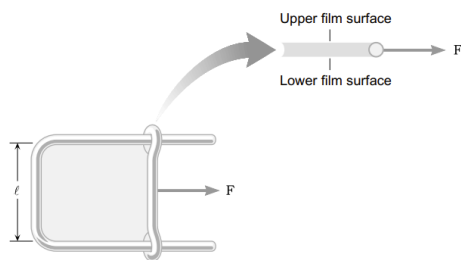
8. Consider steady fully-developed incompressible flow of a Newtonian fluid between two infinite parallel flat plates. The plates move in the opposite directions, as shown in the figure. In the absence of body force and pressure gradient, the ratio of shear stress at the top surface ($y = H$) to that at the bottom surface ($y = 0$) is



9. A skater weighing 1000N skates at a speed of 20m/s on ice maintained at 0 °C. The average skating area supporting the skater is 0.001 m² and the coefficient of friction between the skates and ice is 0.02. What will be the average thickness of a film of water existing at the interface between the skater and ice? ($\mu = 0.01 \text{ N.s/m}^2$)

Surface Tension and Capillary rise

1. A sliding wire of length 3.5 cm is pulling a liquid film, as Figure shows. The pulling force exerted by the wire is 4.4×10^{-3} N. From the table of surface tensions, determine the film material.



Liquid	Surface Tension γ (N/m)
Benzene (20 °C)	0.029
Blood (37 °C)	0.058
Glycerin (20 °C)	0.063
Mercury (20 °C)	0.47
Water (20 °C)	0.073
Water (100 °C)	0.059

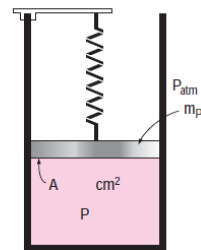
2. A circular ring (radius = 5.0 cm) is used to determine the surface tension of a liquid. The plane of the ring is positioned so that it is parallel to the surface of the liquid. The ring is immersed in the liquid and then pulled upward, so a film is formed between the ring and the liquid. In addition to the ring's weight, an upward force of 3.6×10^{-2} N is required to lift the ring to the point where it just breaks free of the surface. What is the surface tension of the liquid?
3. Suppose that the C-shaped wire frame in above Fig. is rotated clockwise by 90° , so that the sliding wire would fall down if it were free to do so. The sliding wire has a length of 3.5 cm and a mass of 0.20 g. There is no friction between the sliding wire and the vertical sides of the C-shaped wire. If the sliding wire is in equilibrium, what is the surface tension of the film?
4. Two soap bubbles ($\sigma = 0.025$ N/m) have radii of 2.0 and 4.5 mm. For each bubble, determine the difference between the inside and outside pressures.
5. A small bubble of air in water ($\sigma = 0.073$ N/m) has a radius of 0.10 mm. Find the difference in pressures between the inside and outside of the bubble.
6. A drop of oil ($\sigma = 0.0320$ N/m) has a radius of 0.0100 mm. The drop is located a distance of 2.55 m below the surface of fresh water. The atmospheric pressure above the water is 1.01×10^5 Pa. (a) What is the absolute pressure in the water at this depth? (b) Determine the absolute pressure inside the oil drop.
7. Suppose that a bubble has the shape of a long cylinder, rather than that of a sphere. Determine an expression for the difference between the inside and outside pressures; express your answer in terms of the surface tension γ and the radius R of the cylinder.
8. A needle has a length of 3.2 cm. When placed gently on the surface of the water ($\gamma = 0.073$ N/m) in a glass, this needle will float if it is not too heavy. What is the weight of the heaviest needle that can be used in this demonstration?
9. A 0.03-in-diameter glass tube is inserted into kerosene at 68°F . The contact angle of kerosene with a glass surface is 26° . Determine the capillary rise of kerosene in the tube. *Answer:* 0.65 in
10. 1.9-mm-diameter tube is inserted into an unknown liquid whose density is 960 kg/m³, and it is observed that the liquid rises 5 mm in the tube, making a contact angle of 15° . Determine the surface tension of the liquid.
11. Find out the minimum size of glass tube that can be used to measure water level if the capillary rise in the tube is to be restricted to 2 mm. Consider surface tension of water in contact with air as 0.073575 N/m.

Compressibility and Bulk modulus

1. What does the coefficient of compressibility of a fluid represent? How does it differ from isothermal compressibility?
2. What does the coefficient of volume expansion of a fluid represent? How does it differ from the coefficient of compressibility?
3. Can the coefficient of compressibility of a fluid be negative? How about the coefficient of volume expansion?
4. Water at 1 atm pressure is compressed to 800 atm pressure isothermally. Determine the increase in the density of water. Take the isothermal compressibility of water to be $4.80 \times 10^{-5} \text{ atm}^{-1}$.
5. If we neglect the temperature effect, an empirical pressure-density relation for water is $\frac{p}{p_a} = 3001 \times \left(\frac{\rho}{\rho_a}\right)^7 - 3000$, where subscript a refers to atmospheric conditions. Determine the isothermal bulk modulus of elasticity and compressibility of water at 1, 10 and 100 atmospheric pressures.

Pressure

1. A gas is contained in a vertical, frictionless piston– cylinder device. The piston has a mass of 3.2kg and a cross-sectional area of 35 cm^2 . A compressed spring above the piston exerts a force of 150N on the piston. If the atmospheric pressure is 95kPa, determine the pressure inside the cylinder



6. The system above is open to atmospheric pressure (10^5 Pa) on its right side.
 - a) If $L=120 \text{ cm}$, what is the air pressure in container A?
 - b) Conversely, if $p_A = 135 \text{ kPa}$, what is the length L ? Assume the density of water and mercury are $1,000 \text{ kg/m}^3$ and $13,560 \text{ kg/m}^3$, respectively.

